

# Chapter 5

## Sequential Logic

opposed to combinational logic

### 5.1 Objectives

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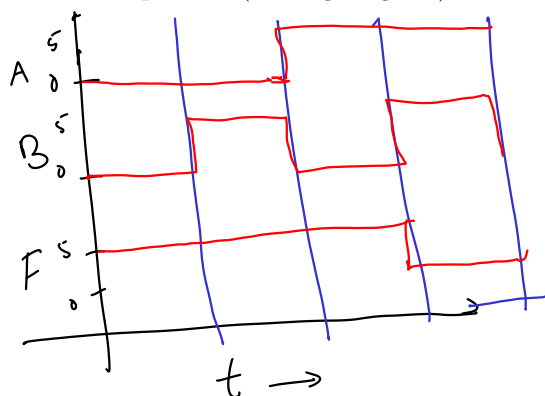
1. Understand timing diagrams, gate delays and critical path
2. Design Hazard-free two level circuits
3. Building blocks of sequential circuits
4. Analyze a sequential circuit and derive a state-table and a state-graph
5. Derive a state graph or state table from a word description of the problem
6. Understanding the structure of an FPGA

### 5.2 Why do we need sequential circuits?

**Example 5.1.** Think about this problem: Design an occupancy counter that depends on a sensor  $S$  at the class door. The sensor is triggered every time a person passes through the door. The counter can be reset to zero with a reset button. Assume we only need up to two bit counter  $C_1C_0$ . Draw a truth table for this circuit. Do you have requisite knowledge for designing this circuit? Can this circuit be designed without a memory element?

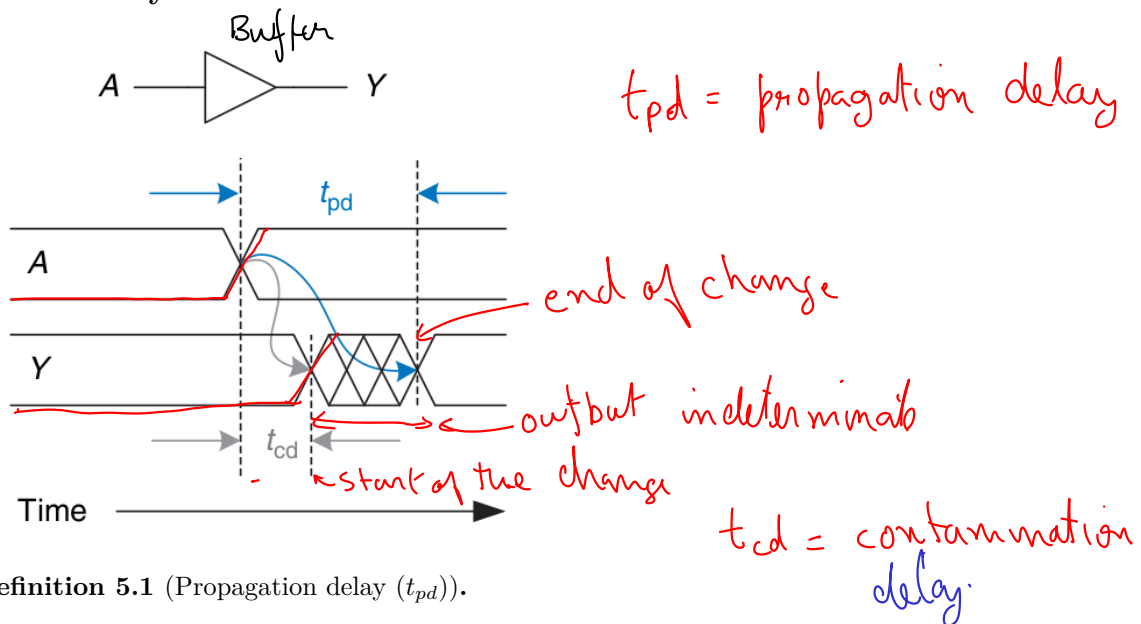
### 5.3 Timing diagrams and propagation delays

**Example 5.2** (Timing diagram). Draw a timing diagram for an ideal NAND gate.



| A | B | F |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

## 5.3.1 Delays



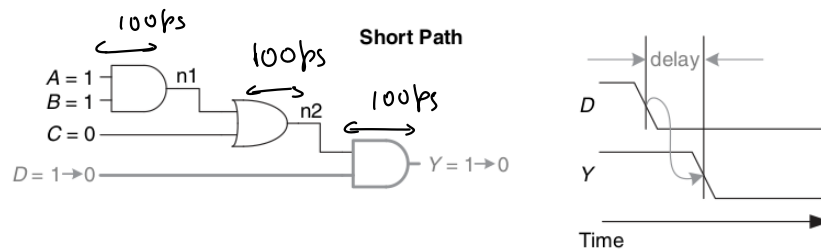
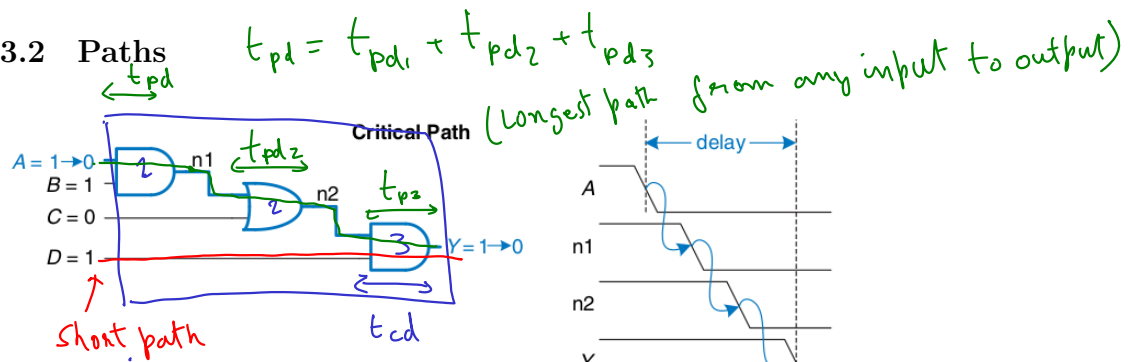
**Definition 5.1** (Propagation delay ( $t_{pd}$ )).

The minimum time it takes for the output signal to stabilize after the input changes.

**Definition 5.2** (Contamination delay ( $t_{cd}$ )).

The time it takes to modify the output signal after change in input

## 5.3.2 Paths



**Example 5.3.** Find the propagation delay of the circuit above given that propagation delay of each gate is 100ps and contamination delay of 60ps.

**Example 5.4** (Ex 2.43 [1]). Determine the propagation delay and contamination delay of the circuit in Figure 5.1. Use the gate delays given in Table 5.2.

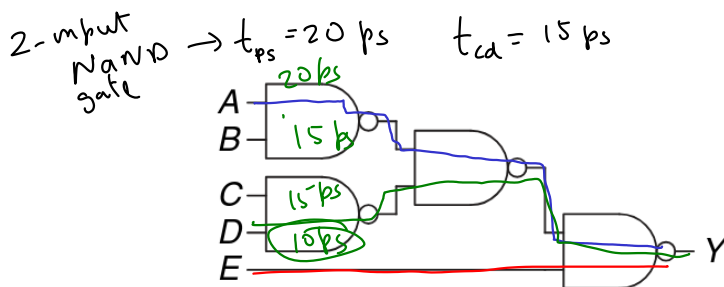


Figure 2.83 Circuit schematic

Figure 5.1: Circuit

Table 2.8 Gate delays for Exercises 2.43–2.47

| Gate         | $t_{pd}$ (ps) | $t_{cd}$ (ps) |
|--------------|---------------|---------------|
| NOT          | 15            | 10            |
| 2-input NAND | 20            | 15            |
| 3-input NAND | 30            | 25            |
| 2-input NOR  | 30            | 25            |
| 3-input NOR  | 45            | 35            |
| 2-input AND  | 30            | 25            |
| 3-input AND  | 40            | 30            |
| 2-input OR   | 40            | 30            |
| 3-input OR   | 55            | 45            |
| 2-input XOR  | 60            | 40            |

Figure 5.2: Delays

**Problem 5.1** (10 marks, Ex 2.44 [1]). *Determine the propagation delay and contamination delay of the circuit in Figure 5.3. Use the gate delays given in Figure 5.2.*

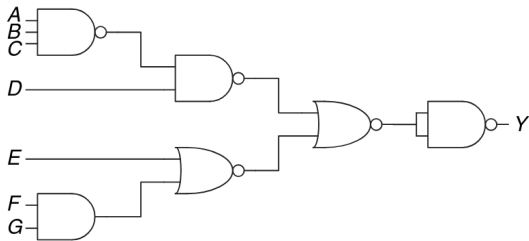
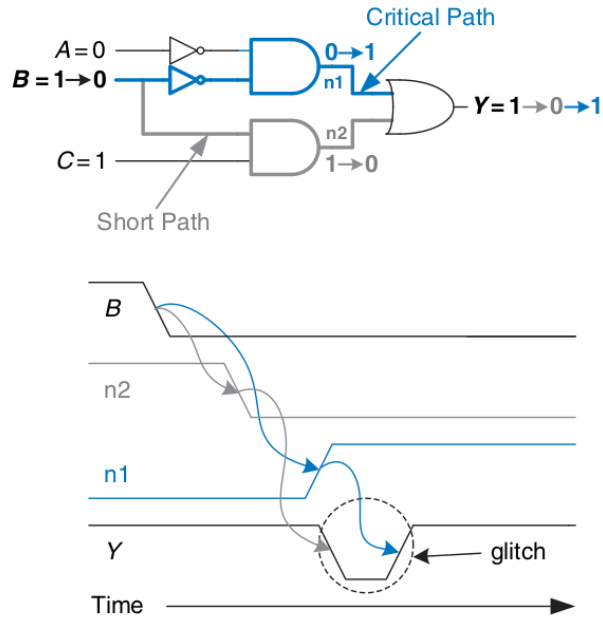


Figure 2.84 Circuit schematic

Figure 5.3: Circuit

## 5.4 Glitches or Hazards



**Definition 5.3** (Glitch or Hazard).

**Example 5.5.** Design a circuit that fixes the glitch in the above circuit (also known as *glitch-free* or *hazard-free* circuit).

**Problem 5.2.** Find a minimal Boolean equation for the function in Figure 2.85. Remember to take advantage of the don't care entries (marked X) (10 marks).

| A | B | C | D | Y |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | X |
| 0 | 0 | 0 | 1 | X |
| 0 | 0 | 1 | 0 | X |
| 0 | 0 | 1 | 1 | 0 |
| 0 | 1 | 0 | 0 | 0 |
| 0 | 1 | 0 | 1 | X |
| 0 | 1 | 1 | 0 | 0 |
| 0 | 1 | 1 | 1 | X |
| 1 | 0 | 0 | 0 | 1 |
| 1 | 0 | 0 | 1 | 0 |
| 1 | 0 | 1 | 0 | X |
| 1 | 0 | 1 | 1 | 1 |
| 1 | 1 | 0 | 0 | 1 |
| 1 | 1 | 0 | 1 | 1 |
| 1 | 1 | 1 | 0 | X |
| 1 | 1 | 1 | 1 | 1 |

Figure 2.85 Truth table for Exercise 2.28

1. Sketch a circuit for the function (10 marks).
2. Does your circuit from have any potential glitches when one of the inputs changes? If not, explain why not. If so, show how to modify the circuit to eliminate the glitches (10 marks).

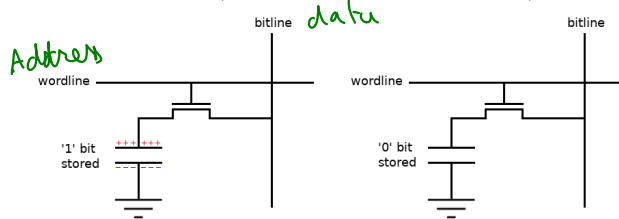
## 5.5 How to create memory element from circuits

Two types of memory

1. Volatile memory. For example, RAM, CPU registers.
2. Non-volatile memory. For example, SSD, Flash drives. (Not covered in this course)

volatile

- (a) Memories that require periodic refreshing. For example, DRAM: Dynamics Random Access memory (Not covered in this course)



1

- (b) Memories that are always refreshing. For example, SRAM: Static Random Access memory [3, Appendix B.64]

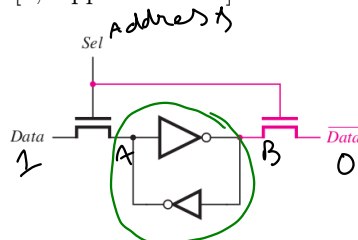
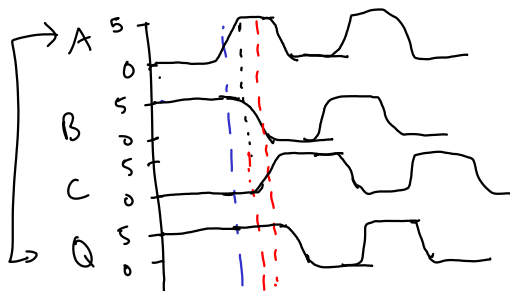
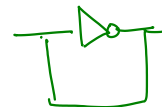
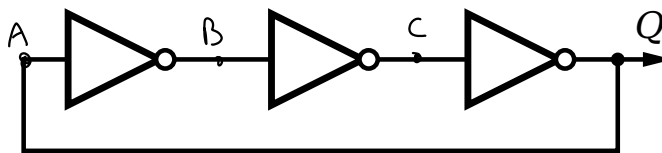


Figure B.64 An SRAM cell.

<sup>1</sup>Image source: [allaboutcircuits.com/technical-articles/introduction-to-dram-dynamic-random-access-memory/](http://allaboutcircuits.com/technical-articles/introduction-to-dram-dynamic-random-access-memory/)

## 5.6 Latches and Flip-Flops [1, Sec 3.2]

**Example 5.6** (Ring oscillator). [1, Sec 3.31] How many stable states does the following circuit have?



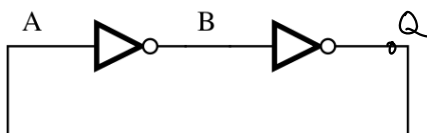
Ring oscillator

Astable circuit because  
this has no stable  
modes of operation  
states

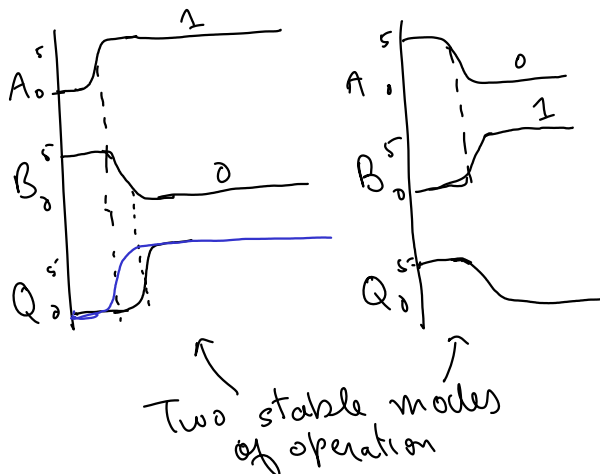
**Definition 5.4** (Astable circuits).

Astable circuits have no stable "modes of operation" or states.

**Example 5.7.** Analyze the timing diagram of the following circuit.



Bistable circuit  
2 states  
1 bit of information



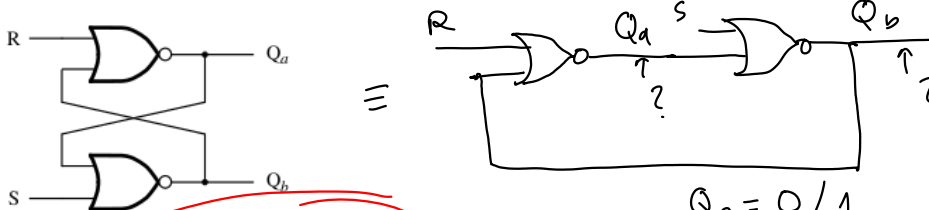
Two stable modes  
of operation

**Definition 5.5** (Bistable circuits).

**Definition 5.6** (Characteristic or state table). Draw the characteristic or state table of the above circuit.

### 5.6.1 SR (Set-Reset) latch [1, Sec 3.2.1]

**Definition 5.7** (SR latch). The following circuit is called the SR latch.



Set =  $S = 1$   
Reset =  $R = 0$

$$Q_a = 0/1$$

$$Q_b = \overline{S + Q_a} = 0$$

$$Q_a = \overline{R + Q_b} = \overline{0 + 0} = 1$$

Set =  $S = 0$   
Reset =  $R = 1$

$$Q_a = \overline{R + Q_b}$$

$$= \overline{1 + Q_b} = \overline{1} = 0$$

$$Q_b = \overline{S + Q_a} = \overline{0 + 0} = 1$$

- 2
1. How many stable states does this circuit have?
  2. Draw its characteristic or state table.
  3. Draw SR latch symbol

|         | S | R | $Q_a^+$ | $Q_b^+$ |
|---------|---|---|---------|---------|
| Hold    | 0 | 0 | $Q_a$   | $Q_b$   |
| Reset   | 0 | 1 | 0       | 1       |
| Set     | 1 | 0 | 1       | 0       |
| Invalid | 1 | 1 | 0       | 0       |

$S = 0$   
 $R = 0$

$$Q_a(t + \Delta t_{pd}) = \overline{R + Q_b(t)}$$

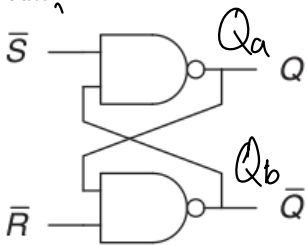
$$= \overline{0 + Q_b(t)} = \overline{Q_b(t)}$$

$$Q_b(t + \Delta t_{pd}) = \overline{S + Q_a(t)} = \overline{Q_a(t)}$$

$$Q_a = 1, Q_b = 0$$

$$Q_a = 1, Q_b = 0$$

**Problem 5.3** (SR latch using NAND gates). Draw the characteristic or state table for the following circuit.



Case 1, 2, 3, 4  
 $S = 0, R = 0$

$S = 0, R = 0$

$$Q_a(t + \Delta t_{pd}) = \overline{S \cdot Q_b(t)}$$

$$= \overline{1 \cdot Q_b(t)}$$

$$= \overline{Q_b(t)}$$

$$Q_b(t + \Delta t_{pd}) = \overline{R \cdot Q_a(t)}$$

| S | R | $Q_a^+$ | $Q_b^+$ |
|---|---|---------|---------|
| 0 | 0 |         |         |
| 0 | 1 |         |         |
| 1 | 0 |         |         |
| 1 | 1 |         |         |

contains  
Q



$$S = 1$$

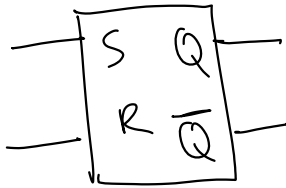
$$R = 1$$

$$Q_a(t + \Delta t_{pd}) = \overline{R + Q_b(t)}$$

$$= \overline{1 + Q_b(t)} = 0$$

$$Q_b(t + \Delta t_{pd}) = \overline{S + Q_a(t)}$$

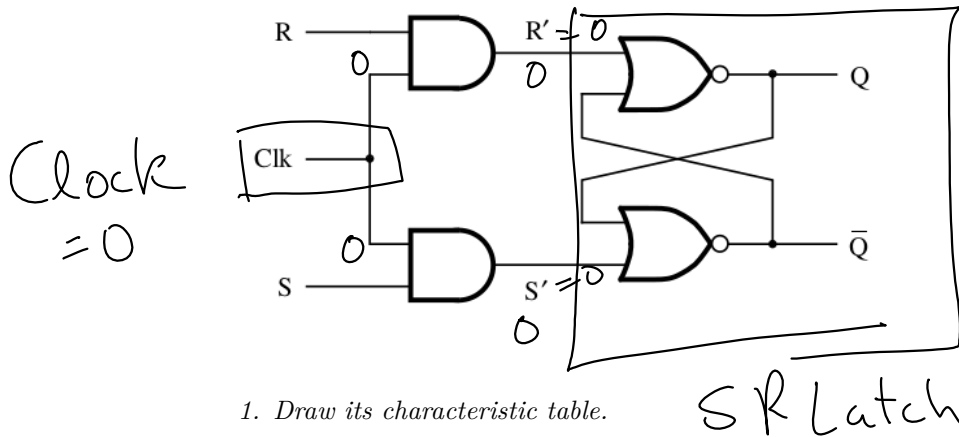
$$= \overline{1 + Q_a(t)} = 0$$



SR Latch

### 5.6.2 Gated SR latch [3, Sec 5.2]

**Definition 5.8** (Gated SR latch). The following circuit is called the Gated SR latch.



1. Draw its characteristic table.

2. Draw the Gated SR latch symbol

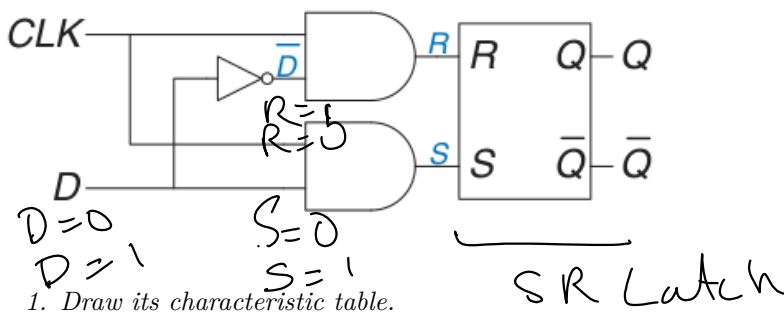
| Clk | S | R | $Q^+$ |         |
|-----|---|---|-------|---------|
| 0   | * | * | Q     | Hold    |
| 1   | 0 | 0 | Q     | Hold    |
| 1   | 0 | 1 | 0     | Reset   |
| 1   | 1 | 0 | 1     | Set     |
| 1   | 1 | 1 | 0     | Invalid |

Characteristic table

| S' | R' | $Q^+$ |         |
|----|----|-------|---------|
| 0  | 0  | Q     | Hold    |
| 0  | 1  | 0     | Reset   |
| 1  | 0  | 1     | Set     |
| 1  | 1  | 0     | Invalid |

### 5.6.3 D (Data) latch [1, Sec 3.2.2]

**Definition 5.9** (D latch). The following circuit is called the D latch.



1. Draw its characteristic table.

2. Draw the D latch symbol

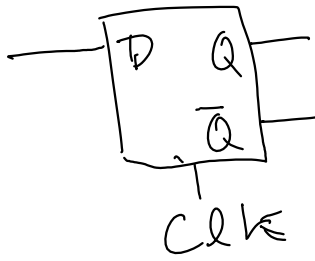
| Clk | D | $Q^+$ |
|-----|---|-------|
| 0   | * | Q     |
| 1   | 0 | 0     |
| 1   | 1 | 1     |

Transparent Latch

| Clk | $Q^+$ |
|-----|-------|
| 0   | Q     |
| 1   | D     |

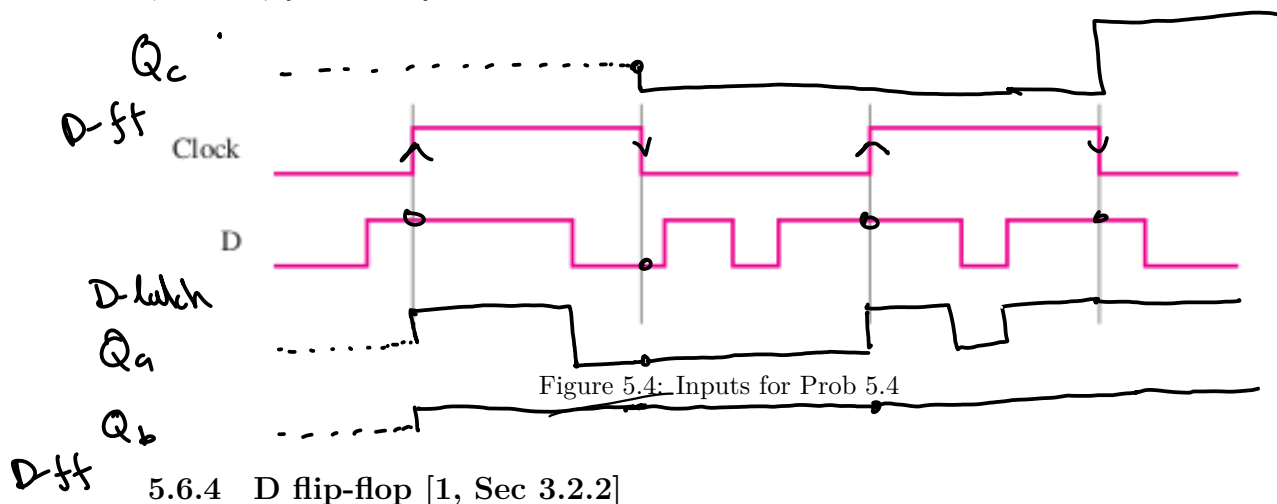
Memory

Hold state  
Reset  
Set



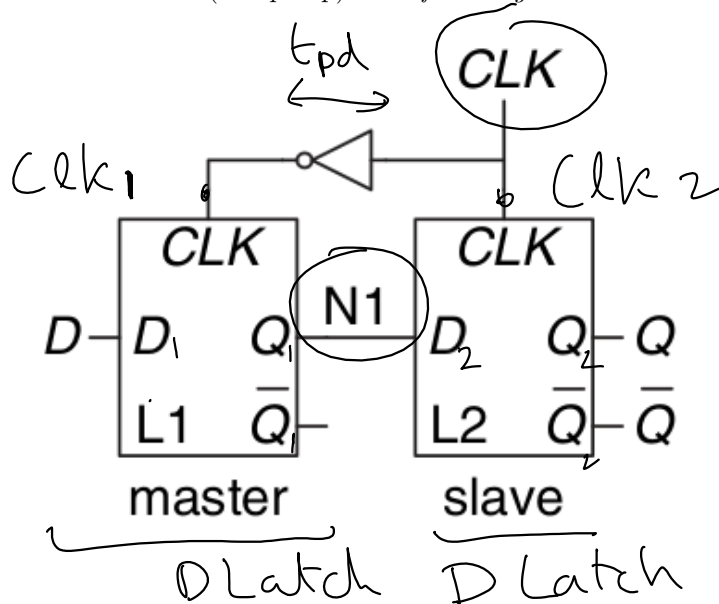
D-Latch Symbol

**Problem 5.4.** Consider the timing diagram in Figure 5.4. Assuming that the  $D$  and Clock inputs shown are applied to the circuit in Figure 5.5, draw waveforms for the  $Q_a$ ,  $Q_b$ , and  $Q_c$  signals. (10 marks) [?, Prob 5.1]



#### 5.6.4 D flip-flop [1, Sec 3.2.2]

**Definition 5.10** (D flip-flop). The following circuit is called the D flip-flop.



1. Draw its timing diagram

2. Draw its characteristic table.

3. Draw the D flip-flop symbol

clock edges = pos edge  
rising edge

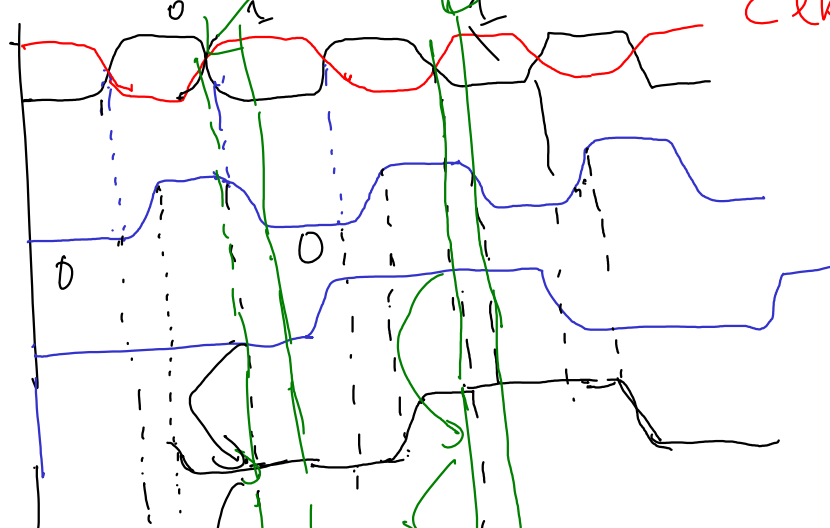
clk2 = CLK

$\overline{\text{CLK2}} = \overline{\text{CLK}}$

CLK1

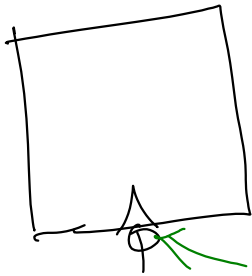
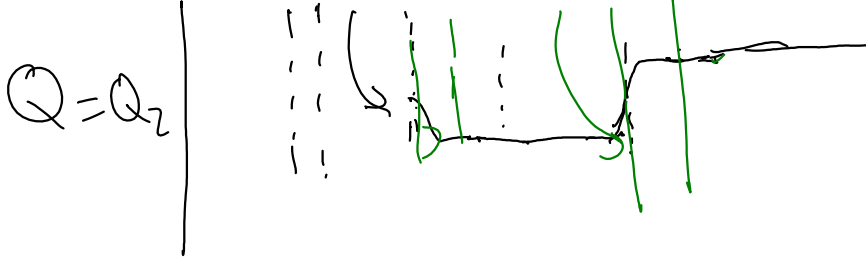
D1

$N1 = Q1$   
 $= D2$

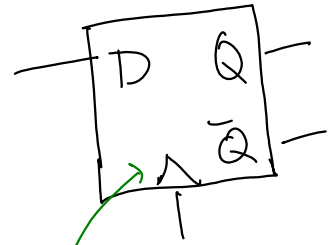


Characteristic  
table

| CLK ↑ | $Q^+$ |
|-------|-------|
| 0     | Q     |
| 1     | D     |



neg edge  
triggered



edge CLK  
triggered

FF = edge  
triggered  
Latch = level  
triggered

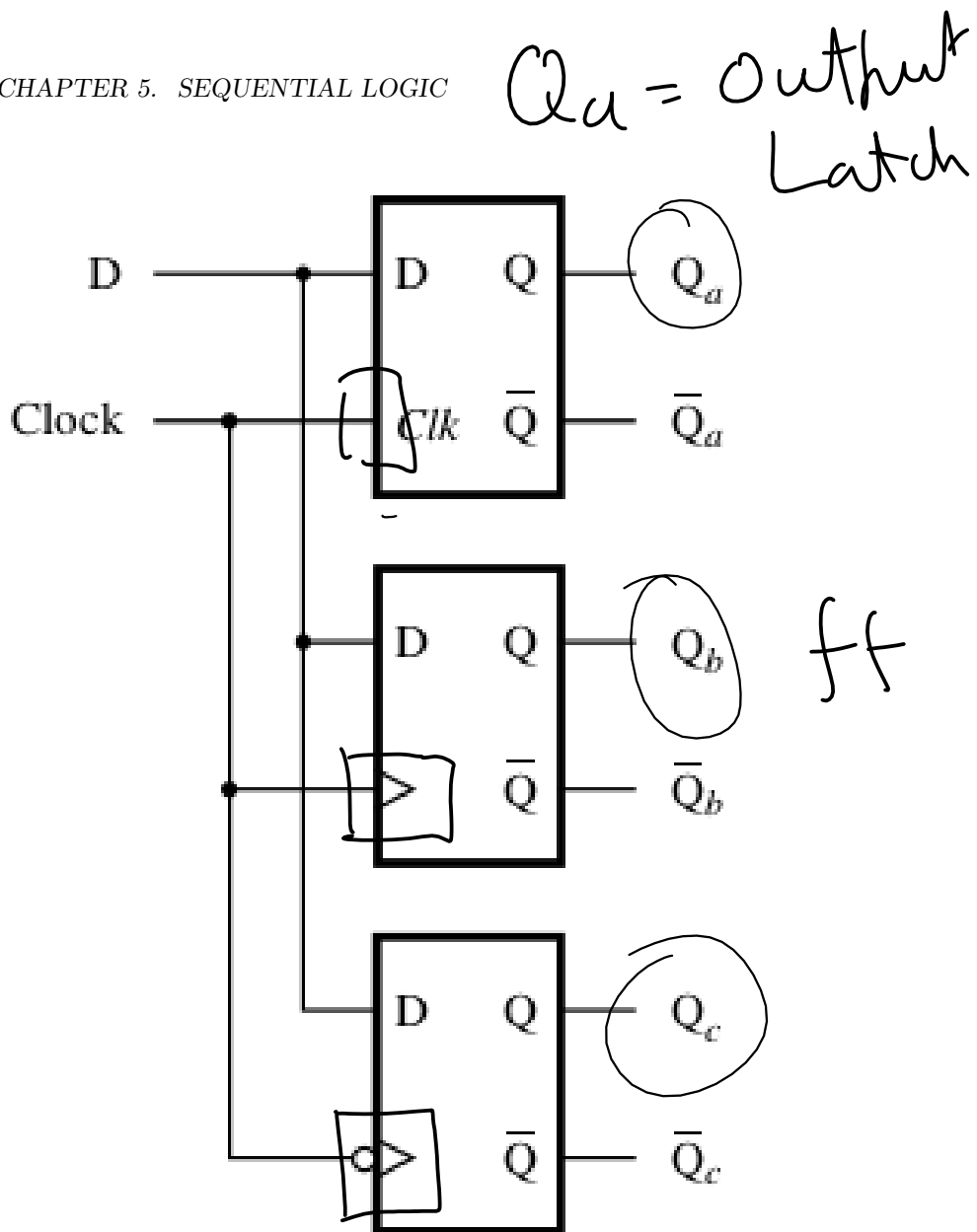
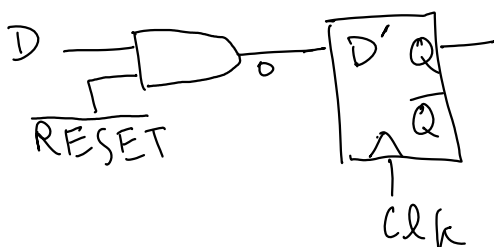


Figure 5.5: Circuit for Prob 5.4. Recall that  $Clk$  with a “▷” symbol indicates rising-edge (positive-edge) triggered flip-flop.  $Clk$  without “▷” symbol indicates a level-triggered latch.  $Clk$  with “◁” symbol indicates a falling-edge (negative-edge) triggered flip-flop.

**Remark 5.1.** What is the difference between a latch and a flip-flop?

**Example 5.8.** Add a RESET signal to the D flip-flop that resets the state of flip-flop to 0.



Synchronous circuit are synchronised with  $clk$  (+ve edge of the clock)

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Asynchronous Reset signal

**Example 5.9.** The toggle (T) flip-flop has one input, CLK, and one output, Q. On each rising edge of CLK, Q toggles to the complement of its previous value. Draw a schematic for a T flip-flop using a D flip-flop and an inverter.

3 kinds of ffs: D-ff, T-ff, JK ff

Data Toggle

Replaced with Toggle

| D-ff |                | T-ff |                | JK ff |   |                |
|------|----------------|------|----------------|-------|---|----------------|
| clk  | Q <sup>+</sup> | T    | Q <sup>+</sup> | J     | K | Q <sup>+</sup> |
| 0    | Q              | 0    | Q              | 0     | 0 | Q              |
| 1    | D              | 1    | $\overline{Q}$ | 0     | 1 | 0              |
|      |                |      |                | 1     | 0 | 1              |
|      |                |      |                | 1     | 1 | $\overline{Q}$ |

Hold Toggle

**Problem 5.5.** A JK flip-flop receives a clock and two inputs, J and K. On the rising edge of the clock, it updates the output, Q. If J and K are both 0, Q retains its old value. If only J is 1, Q becomes 1. If only K is 1, Q becomes 0. If both J and K are 1, Q becomes the opposite of its present state.

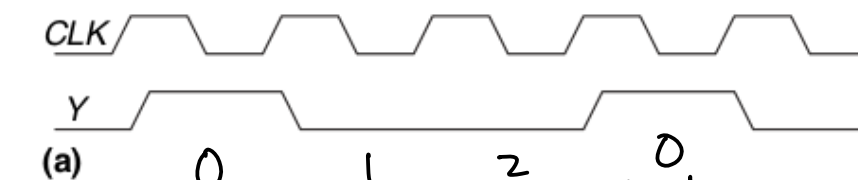
1. Construct a JK flip-flop using a D flip-flop and some combinational logic.
2. Construct a D flip-flop using a JK flip-flop and some combinational logic.
3. Construct a T flip-flop (see Exercise 3.9) using a JK flip-flop.

## 5.7 Finite State Machines [1, Sec 3.4]

2

**Example 5.10.** Design an occupancy counter that depends on a sensor S at the class door. The sensor is triggered every time a person passes through the door. Assume that the counter starts at zero. Assume we only need up to two bit counter  $C_1C_0$ . Draw a state table for this circuit.

**Problem 5.6.** A divide-by-N counter has one output and no inputs. The output Y is HIGH for one clock cycle out of every N. In other words, the output divides the frequency of the clock by N. The waveform for a divide-by-3 counter is shown here:



Sketch circuit designs for such a counter

<sup>2</sup>These notes will not fit on your note sheet.

$$2^1 < 3 \leq 2^2$$

| State     | Next State    | Output |
|-----------|---------------|--------|
| $Q_1 Q_0$ | $Q_1^+ Q_0^+$ | Y      |
| 0 0       | 0 1           | 0      |
| 0 1       | 1 0           | 0      |
| 1 0       | 1 1           | 1      |
| 1 1       | 0 0           | 0      |

# State transition table

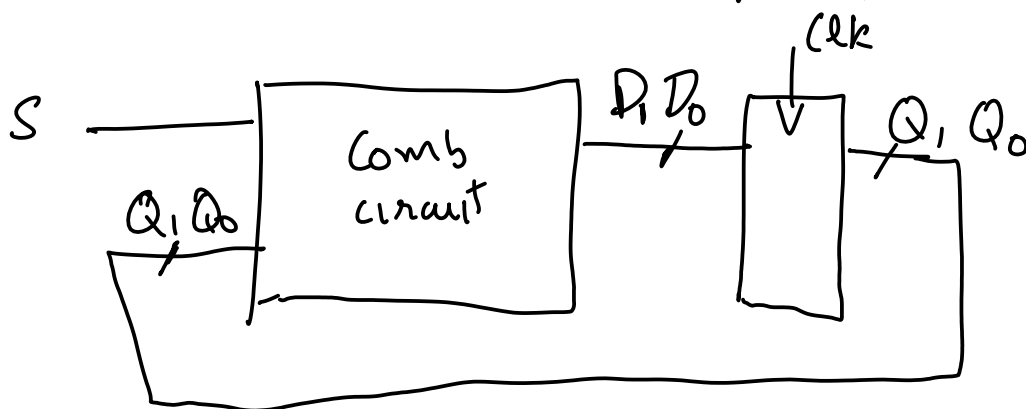
| Input<br>$S$ | State<br>$Q_1 Q_0$ |   | Next State<br>$Q_1^+ Q_0^+$ |   | $D_1$<br>$D_0 \leftarrow$ Truth table |
|--------------|--------------------|---|-----------------------------|---|---------------------------------------|
|              |                    |   |                             |   |                                       |
| 0            | 0                  | 0 | 0                           | 0 |                                       |
| 0            | 0                  | 1 | 0                           | 1 |                                       |
| 0            | 1                  | 0 | 1                           | 0 |                                       |
| 0            | 1                  | 1 | 1                           | 1 |                                       |
| 1            | 0                  | 0 | 0                           | 1 |                                       |
| 1            | 0                  | 1 | 1                           | 0 |                                       |
| 1            | 1                  | 0 | 1                           | 1 |                                       |
| 1            | 1                  | 1 | 0                           | 0 |                                       |

Excitation table

| $D$ | $Q$ | $Q^+$ |
|-----|-----|-------|
| 0   | 0   | 0     |
| 0   | 1   | 0     |
| 1   | 0   | 1     |
| 1   | 1   | 1     |

Design a circuit (Use D-flt whenever not specified)

2-bit Register  $\equiv$  2-flt



$$D_1 D_0 = Q_1^+ Q_0^+$$



$$D(t) = f(Q(t), I(t))$$

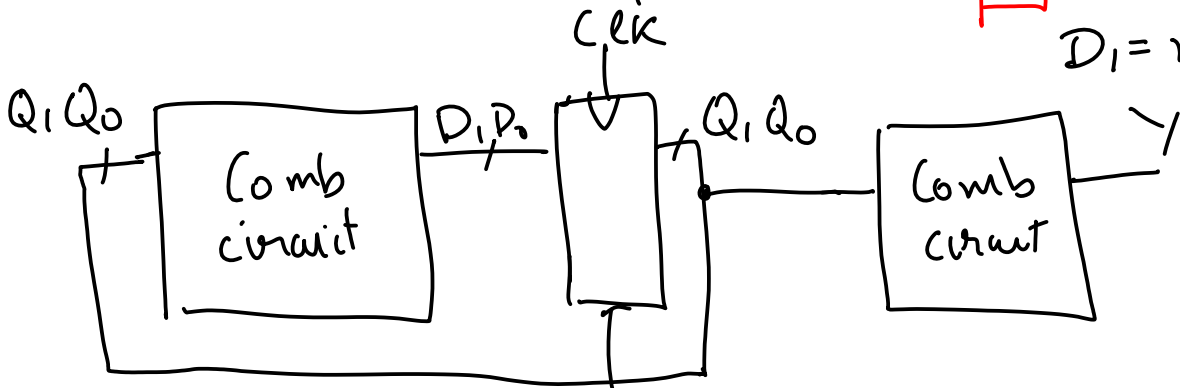
$$D_1 = Q_0 / D_0 = \bar{Q}_1 \bar{Q}_0$$

$$Q(t+1) = D(t)$$

State  
 $Q_1, Q_0$   
 0 0  
 0 1  
 1 0  
 1 1

Next state      Output  
 $Q_1^+, Q_0^+$        $Y$   
 0 1      1  
 0 0      0  
 1 0      0  
 1 1      1

$$D_1 = m_1 = \bar{Q}_1 Q_0$$



| D | Q | Q <sup>+</sup> |
|---|---|----------------|
| 0 | 0 | 0              |
| 0 | 1 | 0              |
| 1 | 0 | 1              |
| 1 | 1 | 1              |

2-bit D-register

$\equiv 2 \text{ D-ff}$

|       | $Q_1$ | $Q_0$ |
|-------|-------|-------|
| $Q_1$ | 0     | 0     |
| $Q_0$ | 1     | 1     |

$$D_1 = Q_0$$

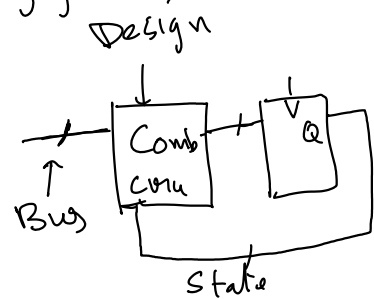
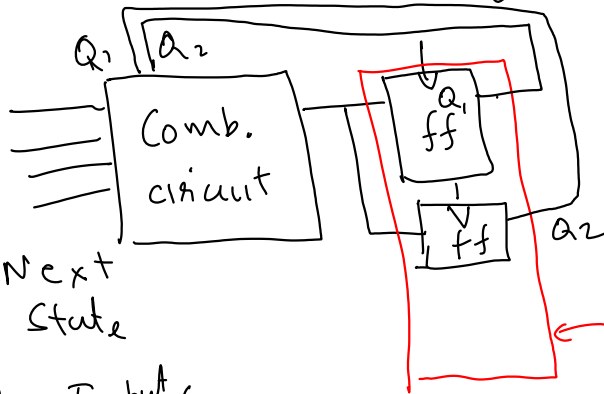
# ① Design of Sequential Circuits (Synchronous)

Design a T-ff using D-ff

Prob

Prev. State → Next State

Inputs + Prev State → Inputs to the Register/ff



A combination of ffs is called register

T-ff from D-ff

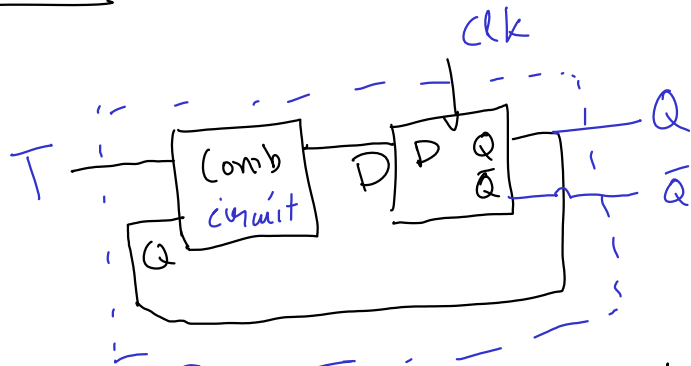
1 bit  
2-states  
1 ff

| T | Q | Q <sup>+</sup> |
|---|---|----------------|
| 0 | 0 | 0              |
| 0 | 1 | 1              |
| 1 | 0 | 1              |
| 1 | 1 | 0              |

State transition table

Truth table

| T | Q | D |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |



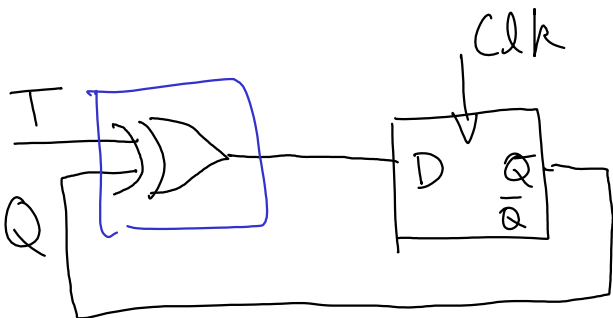
| Q → Q <sup>+</sup> | D |
|--------------------|---|
| 0 → 0              | 0 |
| 0 → 1              | 1 |
| 1 → 0              | 1 |
| 1 → 1              | 0 |

Excitation table

$$D = m_1 + m_2$$

$$= \bar{T}Q + T\bar{Q}$$

$$D = T \oplus Q$$



Practice : Design D-flip using T-flip

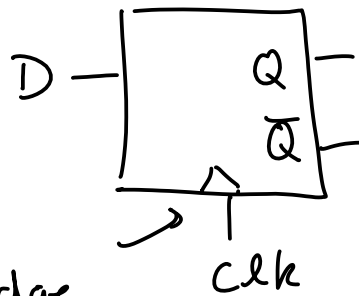
(1) Write State transition table for D-flip

(2) " Excitation table for T-flip

(3) Find Truth table for the comb circuit

# Designing D-ff using J-K ff

D-ff



next state

edge triggered

characteristic table

| D | $Q^+$ |
|---|-------|
| 0 | 0     |
| 1 | 1     |

$Q^+ = D$

memorize

J-K ff

| Set | Reset | $Q^+$     |
|-----|-------|-----------|
| J   | K     |           |
| 0   | 0     | Q         |
| 0   | 1     | 0         |
| 1   | 0     | 1         |
| 1   | 1     | $\bar{Q}$ |

next state

hold

reset

set

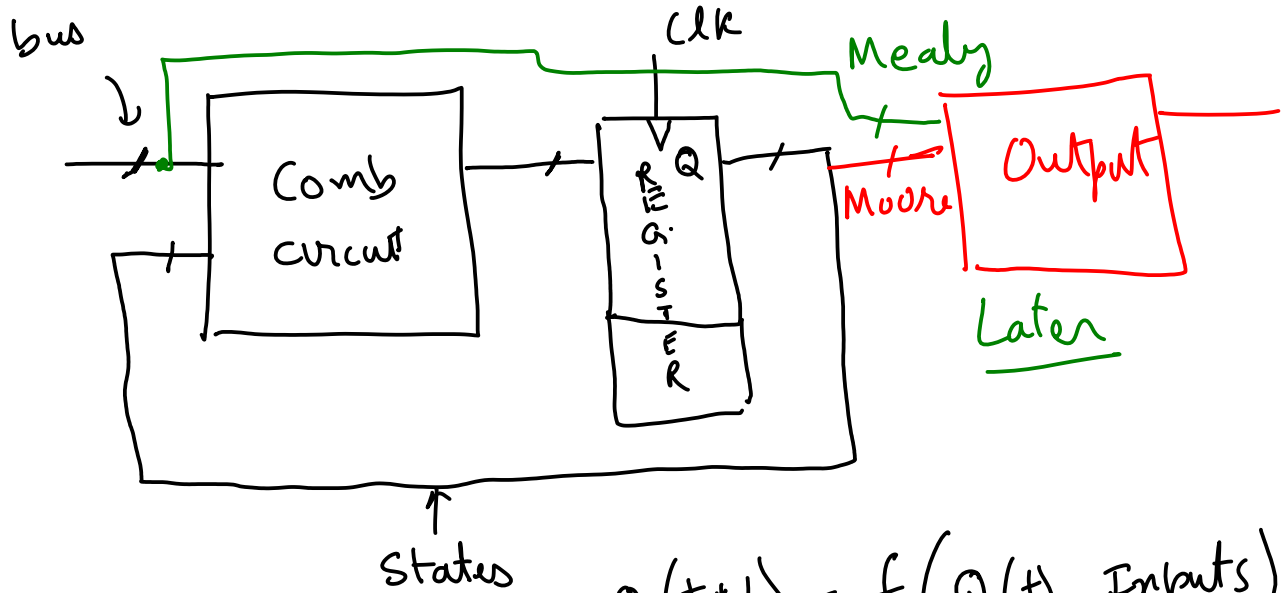
Toggle

Memorize

$$Q^+ = Q(t+1)$$

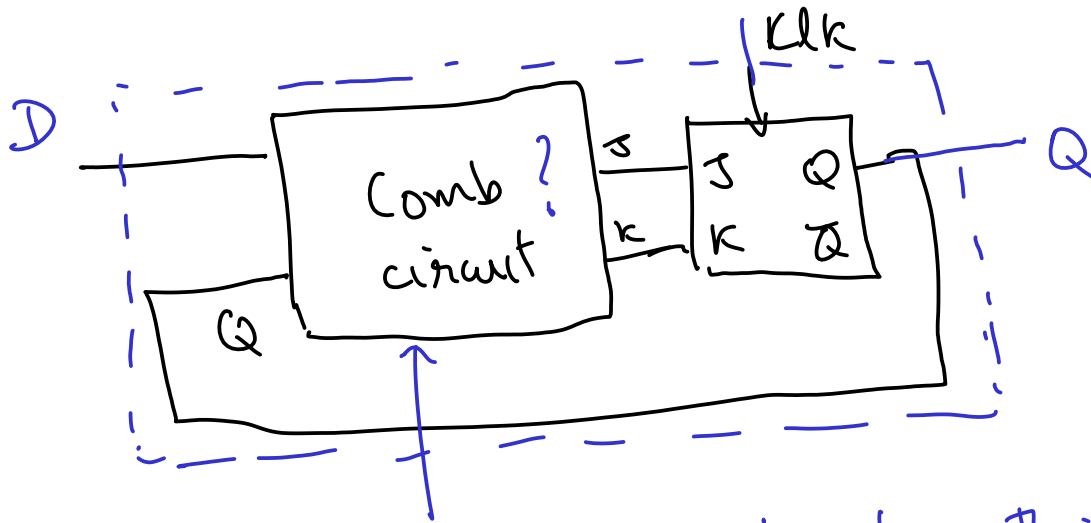
$$Q = Q(t)$$

## Synchronous sequential circuit design



$$Q(t+1) = f(Q(t), \text{Inputs})$$

How many ff? D-ff has 1-bit memory  
 J-K-ff has 1-bit memory  
need 1-ff



find the truth table for this comb circuit

Truth table (find)

| D | Q | J | K |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 1 | 0 | 1 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 1 | 1 |

State Transition table (desired)

| D | Q | Q <sup>+</sup> |
|---|---|----------------|
| 0 | 0 | 0              |
| 0 | 1 | 0              |
| 1 | 0 | 1              |
| 1 | 1 | 1              |

FF excitation table

| Q → Q <sup>+</sup> | J | K |
|--------------------|---|---|
| 0 → 0              | 0 | * |
| 0 → 1              | 1 | * |
| 1 → 0              | * | 1 |
| 1 → 1              | * | 0 |

| J | K | Q <sup>+</sup> |
|---|---|----------------|
| 0 | 0 | Q              |
| 0 | 1 | 0              |
| 1 | 0 | 1              |
| 1 | 1 | Q              |

0 → 0 hold  
 0 → 1 reset  
 1 → 0

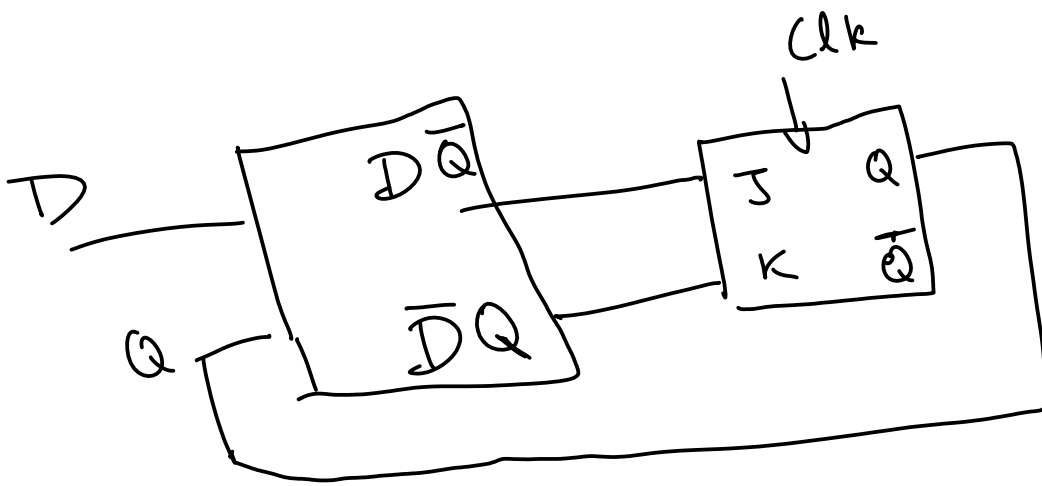
| J | K |
|---|---|
| 0 | 0 |
| 0 | 1 |
| 1 | 0 |
| 1 | 1 |

Q → Q<sup>+</sup>  
 0 → 1 set  
 1 → 1 toggle

what should JK values be for a particular Q-Q<sup>+</sup> transition

$$J = D + Q = D\bar{Q}$$

$$K = \bar{D}Q$$



1. Draw the template circuit
2. Write the state transition table
3. Write the excitation table
4. Use the excitation table to convert state transition table to a truth table
5. Design the combinational circuit from the truth table

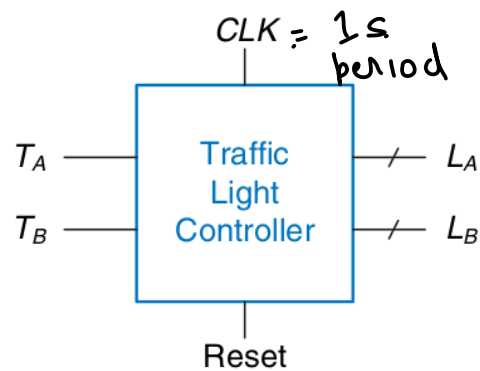
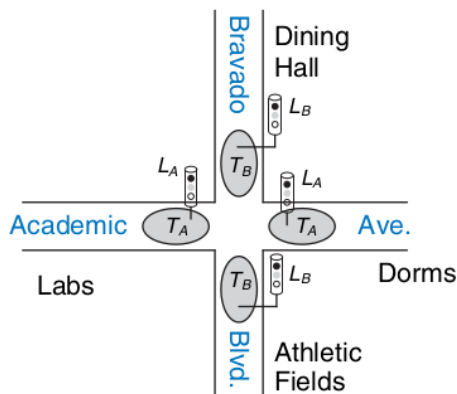
**Problem 5.7.** Design a 3-bit counter which counts in the sequence: 001, 011, 010, 110, 111, 100, (repeat) 001, ...

**Example 5.11.** Design an odd-even counter for an single bit input. The output of this circuit should be 1 if the number of 1s to the input have been odd so far and 0 otherwise.

**Example 5.12** (Sequence detectors). A sequential circuit has one input and one output. The output becomes 1 and remain 1 thereafter when at least two 0's and at least two 1's have occurred as inputs regardless of the order of

**Example 5.13.** Consider the problem of inventing a controller for a traffic light at a busy intersection on campus. There are two traffic sensors,  $T_A$  and  $T_B$ , on Academic Ave. and Bravado Blvd., respectively. Each sensor indicates TRUE if students are present and FALSE if the street is empty. There are two traffic lights,  $L_A$  and  $L_B$ , to control traffic. Each light receives digital inputs specifying whether it should be green, yellow, or red. When the system is reset, the lights are green on Academic Ave. and red on Bravado Blvd. As long as traffic is present on Academic Ave., the lights do not change. When there is no longer traffic on Academic Ave., the light on Academic Ave. becomes yellow for 5 seconds before it turns red and Bravado Blvd.'s light turns green. Similarly, the Bravado Blvd. light remains green as long as traffic is present on the boulevard, then turns yellow and eventually red.

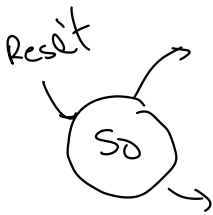
① State transition table/graph



1. Draw a state transition diagram
2. Draw a state table
3. Assign binary encodings to each of the states
4. Redraw the state table with binary encodings. Design a minimal SOP boolean expression.
5. Assign binary encodings to each of the output and redraw the output table. Design a minimal SOP boolean expression for the outputs.

**Problem 5.8.** Design a circuit for a  $2 \times 2$  pixel resolution pong game, where the ball can only occupy 4 possible pixels and a single paddle occupies another 2 pixels. The ball bounces off the paddle when the paddle is in the correct row. To keep it interesting, the ball takes a different path from the source path. Track the score with a single bit counter.

# State graph



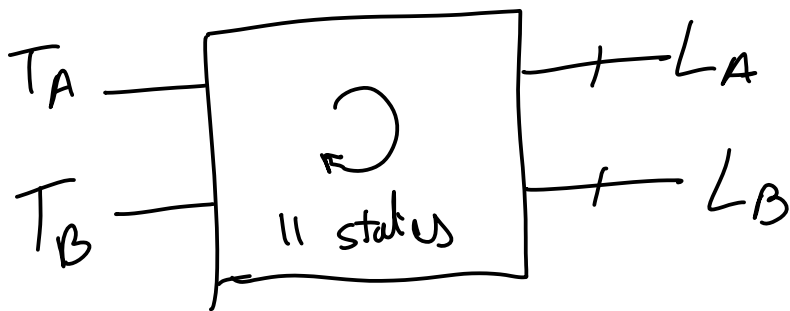
$$\begin{matrix} =00=01 \\ RYG=10 \end{matrix}$$

Red, Yellow, Green

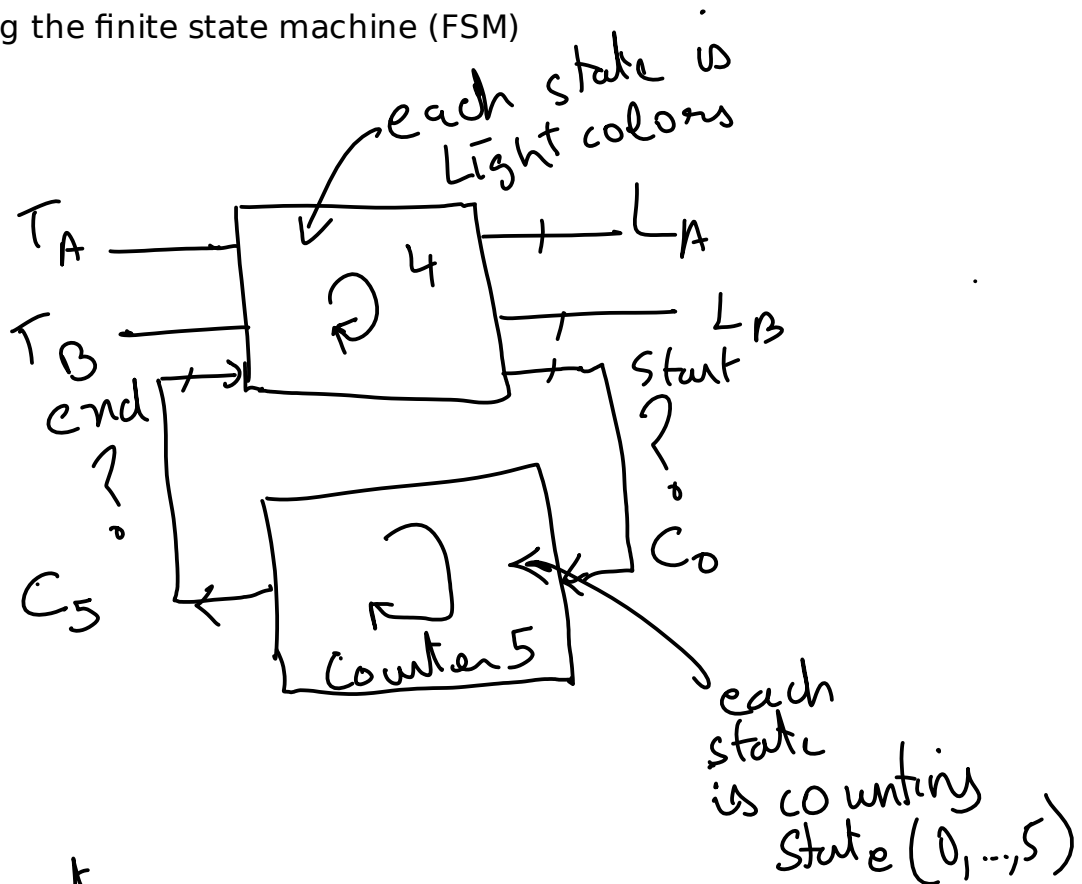
Moore  
Output  
LA LB

| State transition table |                 | Next State        |                 |                 |                 | Moore Output<br>LA LB |   |
|------------------------|-----------------|-------------------|-----------------|-----------------|-----------------|-----------------------|---|
| State mean definition  | Present State   | Inputs<br>TATB=00 | 01              | 10              | 11              |                       |   |
|                        | S <sub>0</sub>  | S <sub>1</sub>    | S <sub>1</sub>  | S <sub>0</sub>  | S <sub>0</sub>  | G                     | R |
|                        | S <sub>1</sub>  | S <sub>2</sub>    | S <sub>2</sub>  | S <sub>2</sub>  | S <sub>2</sub>  | Y                     | R |
| 0                      | S <sub>2</sub>  | S <sub>3</sub>    | S <sub>3</sub>  | S <sub>3</sub>  | S <sub>3</sub>  | Y                     | R |
| 1                      | S <sub>3</sub>  | S <sub>4</sub>    | S <sub>4</sub>  | S <sub>4</sub>  | S <sub>4</sub>  | Y                     | R |
| 2                      | S <sub>4</sub>  | S <sub>5</sub>    | S <sub>5</sub>  | S <sub>5</sub>  | S <sub>5</sub>  | Y                     | R |
| 3                      | S <sub>5</sub>  | S <sub>6</sub>    | S <sub>5</sub>  | S <sub>6</sub>  | S <sub>5</sub>  | R                     | G |
| 4                      | S <sub>6</sub>  | S <sub>7</sub>    | S <sub>7</sub>  | S <sub>7</sub>  | S <sub>7</sub>  | R                     | Y |
|                        | S <sub>7</sub>  | S <sub>8</sub>    | S <sub>8</sub>  | S <sub>8</sub>  | S <sub>8</sub>  | R                     | Y |
|                        | S <sub>8</sub>  | S <sub>9</sub>    | S <sub>9</sub>  | S <sub>9</sub>  | S <sub>9</sub>  | R                     | Y |
| 0                      | S <sub>9</sub>  | S <sub>10</sub>   | S <sub>10</sub> | S <sub>10</sub> | S <sub>10</sub> | R                     | Y |
| 1                      | S <sub>10</sub> | S <sub>0</sub>    | S <sub>0</sub>  | S <sub>0</sub>  | S <sub>0</sub>  | R                     | Y |
| 2                      |                 |                   |                 |                 |                 |                       |   |
| 3                      |                 |                   |                 |                 |                 |                       |   |
| 4                      |                 |                   |                 |                 |                 |                       |   |





Factorizing the finite state machine (FSM)



|    | Present State<br>PS | Next State |    |      |        | Output |    |       |
|----|---------------------|------------|----|------|--------|--------|----|-------|
|    |                     | TA         | TB | = 00 | ... .. | 111    | LA | LB Co |
| C5 |                     |            |    |      |        |        | GR | 0     |
| *  | S0                  |            |    | S1   | S1     | S0 S   | YR | 1     |
| *  | S1                  |            |    | S2   | S2     | S2 S2  | YR | 0     |
| 0  | S2                  |            |    | S2   | S2     | S2 S2  | YR | 0     |
| 1  | S5                  |            |    | S4   | S5     | S6 S5  | RA | 0     |

State transition table  $2^2 \leq 6$  states  $\leq 2^3$  bits

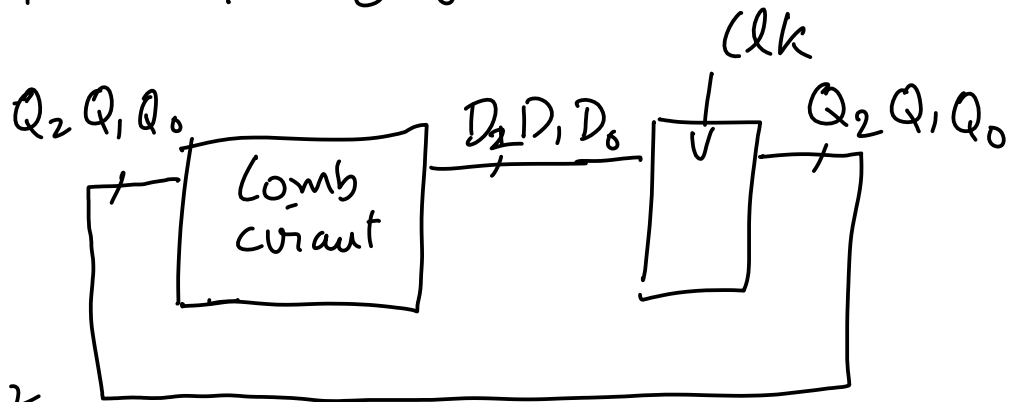
| State |       |       |   | Next State |         |         |
|-------|-------|-------|---|------------|---------|---------|
| $Q_2$ | $Q_1$ | $Q_0$ |   | $Q_2^+$    | $Q_1^+$ | $Q_0^+$ |
| 0     | 0     | 0     |   | d          | d       | d       |
| 0     | 0     | 1     |   | 0          | 1       | 1       |
| 2     | 0     | 1     | 0 | 1          | 1       | 0       |
| 0     | 1     | 1     |   | 0          | 1       | 0       |
| 1     | 0     | 0     |   | 0          | 0       | 1       |
| 1     | 0     | 1     |   | d          | d       | d       |
| 1     | 1     | 0     |   | 1          | 1       | 1       |
| 7     | 1     | 1     | 1 | 1          | 0       | 0       |

Truth Table

$Q_2, Q_1, Q_0$

$$D_2 = m_2 + m_6 + m_7$$

$$D = Q^+$$



| $Q_2$ | $Q_1$ | $Q_0$ | $D_2$ |
|-------|-------|-------|-------|
| d     | 1     | 1     | 0     |
| 0     | 0     | 1     | d     |

$$D_2 = Q_1 \bar{Q}_0 + Q_2 Q_1$$

**Example 1.** A sequential circuit has one input ( $X$ ) and one output ( $Z$ ). The circuit examines groups of four consecutive inputs and produces an output  $Z=1$  if the input sequence 0010 or 0001 occurs. The sequences can overlap. Draw both Mealy and Moore timing diagrams. Find the Mealy and Moore state graph.

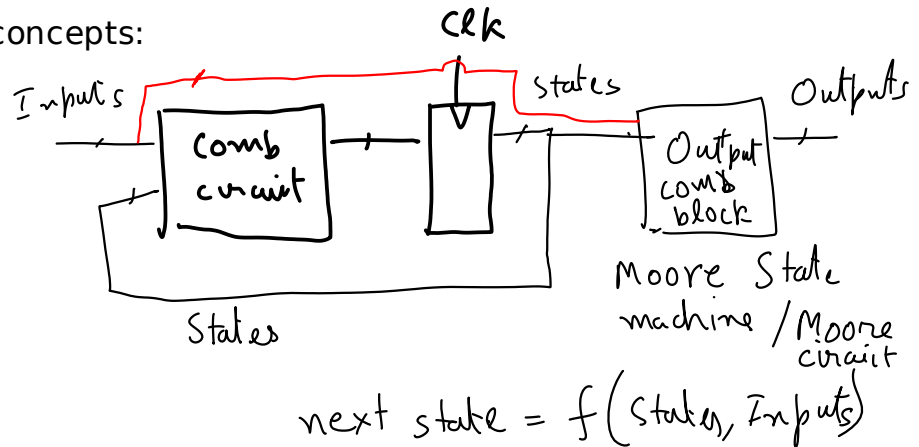
New terminology and concepts:

Mealy:

Moore:

State graph:

State encoding:



Moore state machine/Moore circuit: when output depends only on the states.

Mealy state machine/Mealy circuit: when output depends both on the states and the inputs.

Mealy outputs will change instantaneously when the inputs change.

Moore has the advantage of being synchronized with the clock. For Moore the outputs change only at the rising edge of next clock cycle.

Moore is slower; needs more memory/states/flip flops.

Additional level of complexity:  
State encoding:

binary representation of the State

bits for the state  
 $Q_2 Q_1 Q_0$

|   |   |   |
|---|---|---|
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

Design choice  
Output

Design choice  $\rightarrow$   $\begin{matrix} Q_2 & Q_1 & Q_0 \\ \hline 0 & 0 & 0 \end{matrix} \rightarrow \begin{matrix} Z_2 & Z_1 & Z_0 \\ 0 & 1 & 1 \end{matrix}$

state encoding can be different from output

MOORE circuit (Finite State Machines)

(Finite State Machine + Tape Turing Machine)

Detect 0010 or 0001 with overlap

Clk  
 X 01011100100100 ← Example sequence  
 Z 0000 0 00000100101 ← Desired output

In a Moore circuit the output changes at the next clock cycle after the input changes.

Do not solve for State encoding in the first Step

Use alphabets to denote unique States  $S_0, S_1, S_2, \dots$   
 $A, B, C, D, \dots$

State transition table

Input, State → Next State, Output

X S<sub>0</sub> S<sub>1</sub> 1

Meaning  
 Define  
 state

S<sub>0</sub> = ?  
 S<sub>1</sub> = ?

(moore) State graph

(Mealy) Inputs (on arrows)

Reset

State

output

00010  
 0001

0010 or 0001

00100  
 00101

\*\*\*1  
 \*\*01

Define

S<sub>0</sub> = \*\*\*\*  
 S<sub>1</sub> = \*\*\*0  
 S<sub>2</sub> = \*\*\*1  
 S<sub>3</sub> = \*\*00  
 S<sub>4</sub> = \*001  
 S<sub>5</sub> = 0010  
 S<sub>6</sub> = \*000  
 S<sub>7</sub> = 0001

State graph ≡ State transition table  
 +  
state encoding  
State encoded trans. table

State encoded → optimally assign state encoding  
 k-map based strategies

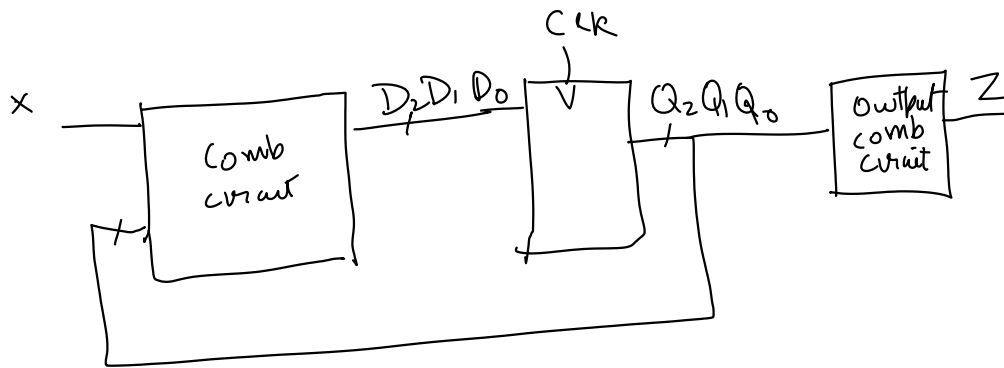
For today just assign any state encoding

# of states = 7 → # of bits / # of ff

$2^2 = 4 < 7 \leq 8 = 2^3$   
 3-bits

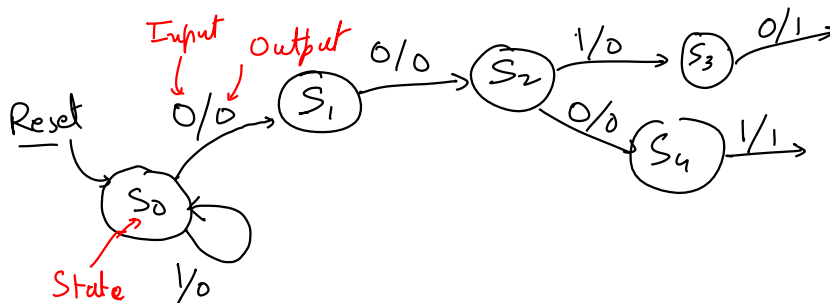
| State | encoding      | Input | State encoded | trans table         | output |
|-------|---------------|-------|---------------|---------------------|--------|
|       | $Q_2 Q_1 Q_0$ | $X$   | $Q_2 Q_1 Q_0$ | $Q_2^+ Q_1^+ Q_0^+$ | $Z$    |
| $S_0$ | 0 0 0         | 0     | 0 0 0         | 0 0 1               | 0      |
| $S_1$ | 0 0 1         | 0     | 0 0 1         |                     |        |
| $S_3$ | 0 1 0         | 0     | 0 1 0         |                     |        |
| $S_4$ | 0 1 1         | 0     | 1 0 0         |                     |        |
| $S_5$ | 1 0 0         | 0     | 1 0 1         |                     |        |
| $S_6$ | 1 0 1         | 0     | 1 1 0         |                     |        |
| $S_7$ | 1 1 0         | 0     | 1 1 1         |                     |        |
|       |               | 1     | 0 0 0         |                     |        |
|       |               | 1     | 0 0 1         |                     |        |
|       |               | 1     | 0 1 0         |                     |        |
|       |               | 1     | 0 1 1         |                     |        |
|       |               | 1     | 1 0 0         |                     |        |
|       |               | 1     | 1 0 1         |                     |        |
|       |               | 1     | 1 1 0         |                     |        |
|       |               | 1     | 1 1 1         |                     |        |

D-flip



1. Define states
2. Draw state graph
3. Assign state encoding
4. Write state encoded trans. table
5. Convert it to Truth table using excitation table
6. Design the required combinational circuits

Mealy State graph Detect 0010 with overlap



Define  
 $S_0 = ****$   
 $S_1 = * * * 0$   
 $S_2 = * * 0 0$   
 $S_3 = * 0 0 1$   
 $S_4 = 0 0 1 0$   
 $S_4 = * 0 0 0$

**Example 5.12** (Sequence detectors). A sequential circuit has one input and one output. The output becomes 1 and remains 1 thereafter when at least two 0's and at least two 1's have occurred as inputs regardless of the order.  
count count .

# Design JK-fl using D-fl

any seq circuit

## ① State transition table

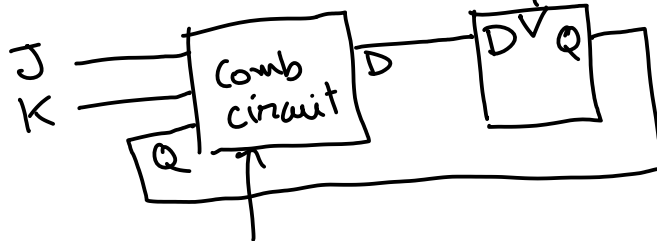
| Inputs |   | Present State | Next State | Outputs |
|--------|---|---------------|------------|---------|
| J      | K | Q             | $Q^+ = D$  | Q       |
| 0      | 0 | 0             | 0          |         |
| 0      | 0 | 1             | 1          |         |
| 0      | 1 | 0             | 0          |         |
| 0      | 1 | 1             | 0          |         |
| 1      | 0 | 0             | 1          |         |
| 1      | 0 | 1             | 1          |         |
| 1      | 1 | 0             | 0          |         |
| 1      | 1 | 1             | 0          |         |

J K  $Q^+$   
0 0 0  
0 0 1  
1 0 0  
1 0 1  
1 1 0  
1 1 1

| J | K | Q | $Q^+$ |
|---|---|---|-------|
| 0 | 0 | 0 | 0     |
| 0 | 0 | 1 | 1     |
| 0 | 1 | 0 | 0     |
| 0 | 1 | 1 | 0     |
| 1 | 0 | 0 | 1     |
| 1 | 0 | 1 | 1     |
| 1 | 1 | 0 | 0     |
| 1 | 1 | 1 | 0     |

$$D = J\bar{Q} + \bar{K}Q$$

Synch Chronous Seq circuit design dentplate



Truth table

| J | K | Q | $D_1$ |
|---|---|---|-------|
| 0 | 0 | 0 |       |
| 0 | 0 | 1 |       |
| 1 | 0 | 0 |       |
| 1 | 0 | 1 |       |
| 1 | 1 | 0 |       |
| 1 | 1 | 1 |       |

D characteristic table

| D | Q | $Q^+$ |
|---|---|-------|
| 0 | 0 | 0     |
| 0 | 1 | 0     |
| 1 | 0 | 1     |
| 1 | 1 | 1     |

$$Q^+ = D$$

Excitation table

| $Q \rightarrow Q^+$ | D |
|---------------------|---|
| 0 → 0               | 0 |
| 0 → 1               | 1 |
| 1 → 0               | 0 |
| 1 → 1               | 1 |

$$D = Q^+$$

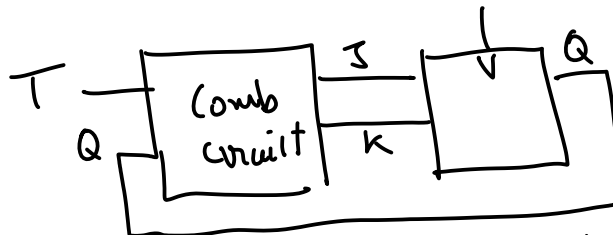
# T-flip using JK-flip

## ① State transition table

| Input | Present State | Next State     | Output |
|-------|---------------|----------------|--------|
| T     | Q             | Q <sup>+</sup> | Q      |
| 0     | 0             | 0              |        |
| 0     | 1             | 1              |        |
| 1     | 0             | 1              |        |
| 1     | 1             | 0              |        |

## ② Truth table

| T | Q | J | K |
|---|---|---|---|
|   |   |   |   |



| J | K | Q <sup>+</sup> |
|---|---|----------------|
| 0 | 0 | Q hold         |
| 0 | 1 | 0 reset        |
| 1 | 0 | 1 set          |
| 1 | 1 | Q̄             |

| Q → Q <sup>+</sup> | J | K |
|--------------------|---|---|
| 0 → 0              | 0 | d |
| 0 → 1              | 1 | d |
| 1 → 0              | d | 1 |
| 1 → 1              | d | 0 |

hold  
reset  
toggle  
set

characteristic table

Excitation table

| Q <sup>+</sup> | T |
|----------------|---|
| 0              | 0 |
| 1              | 1 |

if Q=0: J=Q<sup>+</sup>  
if Q=1: J=d

| Q | T |
|---|---|
| 0 | 0 |
| 1 | 1 |

⇒ J=T

if Q=0: K=d  
if Q=1: K=Q̄

| Q | T |
|---|---|
| 0 | d |
| 1 | 1 |

⇒ K=T